Generative AI for 3D Defect Synthesis, Physics-Aware Modeling, and Calibrated Inspection

Postdoctoral Research Plan and Statement of Research Interests

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1. Research Vision: Physics-Grounded Generative AI for Volumetric Inspection

Non-destructive testing (NDT) and AI-assisted quality control across advanced materials—such as solid-state battery electrodes, composite aerospace structures, and micro-electronics—are bottlenecked by an intrinsic paradox: catastrophic structural defects (micro-cracks, void nucleation, porosity, dendritic growth) are statistically rare in operational data, yet failing to detect them carries severe economic and safety consequences.

Addressing this data scarcity cannot rely on simplistic data augmentation or uncalibrated 2D generative models. It requires **controllable 3D volumetric generative AI** that synthesizes structurally plausible, multi-scale defect morphologies directly into 3D CT, X-ray, and volumetric representations while strictly respecting physical attenuation, material contrast, and reconstruction geometry.

My doctoral research has investigated the theoretical and systems foundations of visual perception under low supervision ($\Delta\theta = 0$), proving the benefits of distributional evidence retention, 2D-to-3D geometric lifting, and probabilistic calibration. At King Abdullah University of Science and Technology (KAUST), I propose to unite these principles to develop physics-aware generative defect synthesis engines that enable robust sim-to-real transfer for industrial inspection.

2. Methodological Foundations from Doctoral Research

My doctoral dissertation, *Visual Perception under Low Supervision: Inference-Time Coordination of Frozen Foundation Model Representations* (supervised by Prof. Ebenezer Owusu), contributes three pillars directly applicable to 3D defect modeling:

2.1 Semantic Retention: Beyond Argmax (ICLR 2027 Submission)

In our recent submission to **ICLR 2027**, titled *Beyond Argmax: A Mechanistic Study of Semantic Retention in Frozen Foundation-Model Composition for Generalized Few-Shot 3D Segmentation*, I identified and mathematically characterized **premature decision collapse**. Standard perceptual pipelines commit prematurely to hard, greedy decisions (argmax), permanently truncating low-probability semantic alternatives. Across 312 3D scenes ($\sim$50M points) of ScanNet200 and ScanNet++, I demonstrated that retaining compact top-k candidate distributions enables downstream geometric consensus to recover rare classes, boosting harmonic mean by +6.40 points. In 3D defect synthesis, preserving continuous candidate distributions across spatial voxels is essential for generating subtle morphological gradations (e.g., crack tip stress concentrations) that reflect genuine material physics.

2.2 3D Geometric Lifting and Multi-View Projection (ACCV 2026, Paper #482)

In ACCV 2026, I constructed training-free geometric lifting architectures that cast camera-pose rays (K, T) to align dense 3D representations with promptable 2D segmenters via spatial consensus. This mathematical machinery maps directly to CT and X-ray imaging: forward projection, Radon transforms, and multi-angle projection geometry, ensuring synthetic 3D defects can be rendered consistently into 2D radiographic projections and reconstructed volumetric slices.

2.3 Reliability Calibration for Industrial Safety (Target: IEEE TPAMI In Prep)

In industrial NDT, false negatives are intolerable. In research in preparation for *IEEE TPAMI*, I formulated **Reliability-Calibrated Adaptive Semantic Arbitration (RCASA)**. By decoupling classifier confidence from contextual reliability through Platt scaling and Bernoulli uncertainty modeling, RCASA guarantees mathematically bounded updates. This provides formal calibration tools to verify that synthetic defect datasets do not distort downstream classifier calibration.

3. Proposed Postdoctoral Research Program at KAUST

Within the Physical Science and Engineering Division at KAUST, I propose to pursue three integrated research objectives:

Theme A: Controllable 3D Generative Defect Synthesis via Diffusion and Implicit Fields

I will develop generative architectures (3D Denoising Diffusion Probabilistic Models and continuous neural implicit representations) designed to synthesize controllable defect geometries (e.g., fatigue cracks, delaminations, gas porosity, dendritic branching in batteries). Rather than inserting rigid geometric templates, the synthesis engine will condition on material composition, mechanical stress tensors, and spatial boundary constraints, generating diverse, organic defect morphologies that match empirical micro-CT observations.

Theme B: Physics-Aware X-ray Attenuation and Forward Projection Modeling

To bridge the sim-to-real domain gap, synthetic defect volumes must obey radiative transfer physics. I propose to integrate differentiable forward projection pipelines (incorporating Beer-Lambert attenuation laws, beam hardening, partial volume effects, and polychromatic X-ray spectra). By training generative models under multi-view radiographic consistency losses, the framework will ensure that synthesized defects produce physically faithful CT reconstruction artifacts when processed through standard FDK or iterative reconstruction algorithms.

Theme C: Calibrated Sim-to-Real Benchmark for Downstream Inspection Pipelines

Synthesizing realistic defects is only valuable if it provably enhances downstream inspection. I will integrate the generated 3D volumetric datasets into standard detection and segmentation workflows (e.g., 3D U-Net, YOLOv8/v10, and zero-shot foundation models). Using my RCASA calibration framework, I will quantify epistemic uncertainty and benchmark the exact synthetic-to-real transfer efficiency across multi-scale CT benchmarks, establishing certified safety margins for automated non-destructive evaluation.

4. Synergies with KAUST Infrastructure and Research Ecosystem

KAUST provides an unparalleled research ecosystem, combining world-leading materials science laboratories, advanced imaging facilities, and the Shaheen III supercomputing platform. My technical proficiencies in PyTorch, CUDA fundamentals, Triton, and memory-bounded streaming architectures will enable me to train high-resolution 3D volumetric models efficiently on large-scale GPU clusters.

Having founded a software consultancy and lectured at Accra Technical University, I combine technical autonomy with proven project leadership. I am deeply eager to collaborate with faculty, postdocs, and graduate researchers in the Physical Science and Engineering Division at KAUST to pioneer generative AI for material science and industrial safety.